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Experiment 6

6. a. Develop an efficient implementation of Floyd's algorithm to determine the shortest paths between all pairs of vertices in weighted graph. Floyd's Algorithm efficiently computes this using dynamic programming. The input should be a weighted adjacency matrix representing the graph, and the output should be the shortest distance matrix showing the shortest path distances between every pair of vertices.

Type of Lab Activity: Exercise

Aim	Material	Methods	Answer
Given	Given	Given as Algorithm	Open

Algorithm

ALGORITHM *Floyd*($W[1..n, 1..n]$)

//Implements Floyd's algorithm for the all-pairs shortest-paths problem

//Input: The weight matrix W of a graph with no negative-length cycle

//Output: The distance matrix of the shortest paths' lengths

$D \leftarrow W$ //is not necessary if W can be overwritten

for $k \leftarrow 1$ **to** n **do**

for $i \leftarrow 1$ **to** n **do**

for $j \leftarrow 1$ **to** n **do**

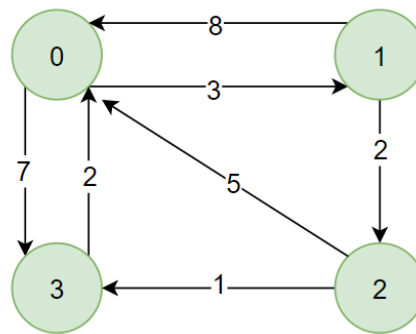
$D[i, j] \leftarrow \min\{D[i, j], D[i, k] + D[k, j]\}$

return D

Program

// Your Program Goes Here//

Sample Output



The following matrix shows the shortest distances between every pair of vertices:

0	3	5	6
5	0	2	3
3	6	0	1
2	5	7	0